BEING UNIVERSALLY CLOSED IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites two Morphisms of Spaces lemmas, one Morphisms lemma and one Properties of Spaces lemma plus same-chapter items, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Framework lemma for descending properties of morphisms. Stacks Project, Descent on Spaces, Lemma lemma-descending-properties-morphisms. Supplies the criterion structure that the proof verifies.
- Universal closedness is \'etale local on the base. Stacks Project, Morphisms of Spaces, Lemma spaces-morphisms-lemma-universally-closed-local. Supplies assumptions (1), (2) and the affine test reduction.
- fpqc quotient topology. Stacks Project, Morphisms, Lemma morphisms-lemma-fpqc-quotient-topology. Shows $|Y \times_Z Z'| \to |Y|$ is submersive.
- Points of fibres of cartesian squares. Stacks Project, Properties of Spaces, Lemma spaces-properties-lemma-points-cartesian. Supplies the surjectivity of the points map used in the final assembly.

Source reference. Original lemma 041J: sources/stacks-chow/tags.txt, source lines 5294–5294. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03IT: sources/stacks-chow/tags.txt, source lines 4615–4615. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original equation 041M: sources/stacks-chow/tags.txt, source lines 5297–5297. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02JY: sources/stacks-chow/tags.txt, source lines 3353–3353. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03H4: sources/stacks-chow/tags.txt, source lines 4553–4553. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. The property $\mathcal{P}(f) =$ "f is universally closed" is fpqc local on the base on algebraic spaces over S.*

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Proof. We will use Lemma 041J to prove this. Assumptions (1) and (2) of that lemma follow from Morphisms of Spaces, Lemma 03IT. Let $Z' \to Z$ be a surjective flat morphism of affine schemes over S. Let $f : X \to Z$ be a morphism of algebraic spaces, and assume that the base change $f' : Z' \times_Z X \to Z'$ is universally closed. We have to show that f is universally closed. To see this, using Morphisms of Spaces, Lemma 03IT again, it is enough to show that for every affine scheme Y and morphism $Y \to Z$ the map $|Y \times_Z X| \to |Y|$ is closed. Consider the cube (041M). The assumption that f' is universally closed implies that $|Y \times_Z Z' \times_Z X| \to |Y \times_Z Z'|$ is closed. As $Y \times_Z Z' \to Y$ is quasi-compact, surjective, and flat as a base change of $Z' \to Z$ we see the map $|Y \times_Z Z'| \to |Y|$ is submersive, see Morphisms, Lemma 02JY. Moreover the map

$$|Y \times_Z Z' \times_Z X| \longrightarrow |Y \times_Z Z'| \times_{|Y|} |Y \times_Z X|$$

is surjective, see Properties of Spaces, Lemma 03H4. It follows by elementary topology that $|Y \times_Z X| \to |Y|$ is closed. □